Chapter 16 — Spring Cloud and Microservices

A **microservice architecture** splits one application into small, independently deployed services talking over the network. It buys independent releases and scaling, and costs every problem of **distributed systems** (parts on different machines over an unreliable network; System Design Fundamentals): where is the other service, what if it is slow, what is the single entry point, where does configuration live, how do you follow one request through ten services. **Spring Cloud** is a family of projects giving Boot applications ready-made answers.

This chapter covers the *Spring implementation*. The patterns — gateways (one front door for all clients), circuit breakers (a switch that stops calls to a failing service), sagas (multi-service workflows of local steps plus undo steps), CQRS (separate write and read models) — are taught in the Microservices Patterns notes and System Design Patterns; each section links there.

The code: four Spring Boot 4.1.1 projects on Java 25 in `code/` — `config-demo`, `service-clients`, `gateway-demo`, `resilience-demo` (7 tests, all run). Downstream services (the ones a service calls) are tiny local HTTP servers on random ports, so nothing external is needed. Outputs are real `mvn -q verify` output (log noise trimmed as `...`).

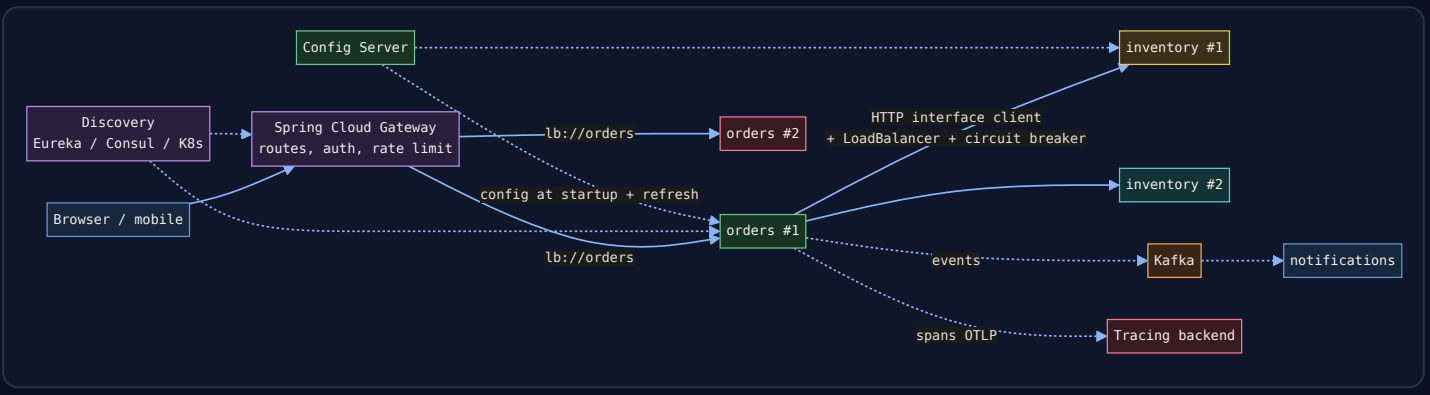
> **Prerequisites.** HTTP clients and interfaces: Chapter 6. Proxies and Framework 7 `@Retryable`: Chapter 4. `spring.config.import`: Chapter 3. Tracing and health probes: Chapter 15. Messaging: Chapter 14.

16.1 The map: microservice problems and Spring answers

Why it matters

Know which problem a dependency solves — and whether your platform already solves it. Many Kubernetes teams add Eureka and a Config Server out of habit, doubling the moving parts.

Problem	Pattern notes	Spring answer	On Kubernetes, often instead
One entry point for clients, cross-cutting edge concerns (auth, rate limits)	API Gateway, BFF	**Spring Cloud Gateway**	Ingress / Gateway API controller, service mesh (sidecar proxies handling routing, retries and encryption between services)
Find instances whose addresses change	Service Discovery	Eureka, Consul, Zookeeper, Spring Cloud Kubernetes	Kubernetes Services + DNS
Spread calls over instances	Load balancing	**Spring Cloud LoadBalancer** (client side)	kube-proxy (forwards a Service's traffic to its pods) / mesh (server side)
Configuration per environment, changed without rebuild	Config externalization	**Spring Cloud Config**, Vault, Consul KV	ConfigMaps / Secrets (Kubernetes config and secret objects; `spring.config.import=configtree:`)
Failing or slow dependencies	Circuit Breaker, Retry, Bulkhead	**Resilience4j** (+ Spring Cloud CircuitBreaker), Framework 7 `@Retryable` / `@ConcurrencyLimit`	mesh retries / outlier detection (ejecting repeatedly failing instances; coarser)
Following a request across services	Distributed Tracing	Micrometer Tracing + OpenTelemetry (15.7)	same, plus mesh telemetry
Declarative HTTP calls	—	HTTP interface clients (Framework), OpenFeign (maintenance)	—
Async communication, events	Event-Driven, Saga, CQRS	Spring Kafka / AMQP (Chapter 14), Spring Cloud Stream	—
Cross-cutting helpers next to each service	Sidecar	—	sidecar containers, service mesh



When not to use microservices

Distributed systems are harder to build, test, debug and operate. A small team, an unclear domain or no need for independent scaling are strong reasons to start with a **well-modularised monolith** (one deployable with clean internal module boundaries; Spring Modulith helps enforce them) and extract services later via Strangler Fig. See DDIA for the consistency problems you inherit.

16.2 Release trains and versions

What it is

Spring Cloud is ~15 projects (Commons, Config, Gateway, OpenFeign, Netflix, Consul, Kubernetes, Circuit Breaker, Stream, Function, Bus, Vault, Task, Zookeeper...), each versioned separately. A **release train** is a BOM (bill of materials) pinning versions that work together and with one Boot generation, named `YYYY.MINOR.PATCH` (since 2020) plus a codename.

Train	Codename	Spring Boot	Component versions
2025.1.x	**Oakwood**	**4.0.x, 4.1.x** (4.1 from 2025.1.2)	5.0.x
2025.0.x	Northfields	3.5.x	4.3.x — open-source support ended June 2026
2024.0.x	Moorgate	3.4.x	4.2.x

(Source: spring.io/projects/spring-cloud compatibility table; 2025.1.3 (August 2026) is the latest service release at the time of writing — based on Boot 4.0.x, compatible with 4.1.x. A `2026.0.0-M1` milestone exists on Maven Central.)

```
<properties>
  <spring-cloud.version>2025.1.3</spring-cloud.version>
</properties>
<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.cloud</groupId>
      <artifactId>spring-cloud-dependencies</artifactId>
      <version>${spring-cloud.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
<!-- then Spring Cloud starters without <version> -->
```

Recent changes that trip people up:

- **Gateway artifacts were renamed** (2025.0): `spring-cloud-starter-gateway-server-webflux` (reactive) and `spring-cloud-starter-gateway-server-webmvc` (servlet); properties moved to `spring.cloud.gateway.server.webflux.*` / `...server.webmvc.*`.
- **Spring Cloud Sleuth is gone** (last in 2021.0 for Boot 2.x); tracing is Micrometer Tracing in Boot itself.
- **Netflix Hystrix, Ribbon, Zuul** were removed years ago; use Resilience4j, Spring Cloud LoadBalancer and Spring Cloud Gateway.
- **OpenFeign** is feature-complete (maintenance mode) — the Spring team recommends HTTP interface clients.
- **Spring Cloud Contract**: the 2025.1.0–2025.1.2 BOMs manage it (5.0.x), but the 2025.1.3 BOM no longer lists it, and its GitHub repository has been moved to the `spring-attic` organisation and archived (read-only; last commit July 2026). Existing users can pin `spring-cloud-contract-dependencies` 5.0.3 explicitly; for new projects consider consumer-driven contracts with Pact (16.10).
- Resilience4j is **not** managed by Spring Boot; the Spring Cloud CircuitBreaker BOM pins Resilience4j 2.3.0, while Resilience4j's own Boot 4 module is `resilience4j-spring-boot4` 2.4.0 (used standalone in 16.8).

Common mistakes

- Mixing trains and Boot generations (e.g. 2025.0 with Boot 4) → `NoSuchMethodError` / `ClassNotFoundException`. Spring Cloud checks compatibility at startup and fails fast with a clear message (`spring.cloud.compatibility-verifier.enabled`).
- Hard-coding component versions instead of importing the BOM.

16.3 Centralized configuration: Spring Cloud Config

What it is

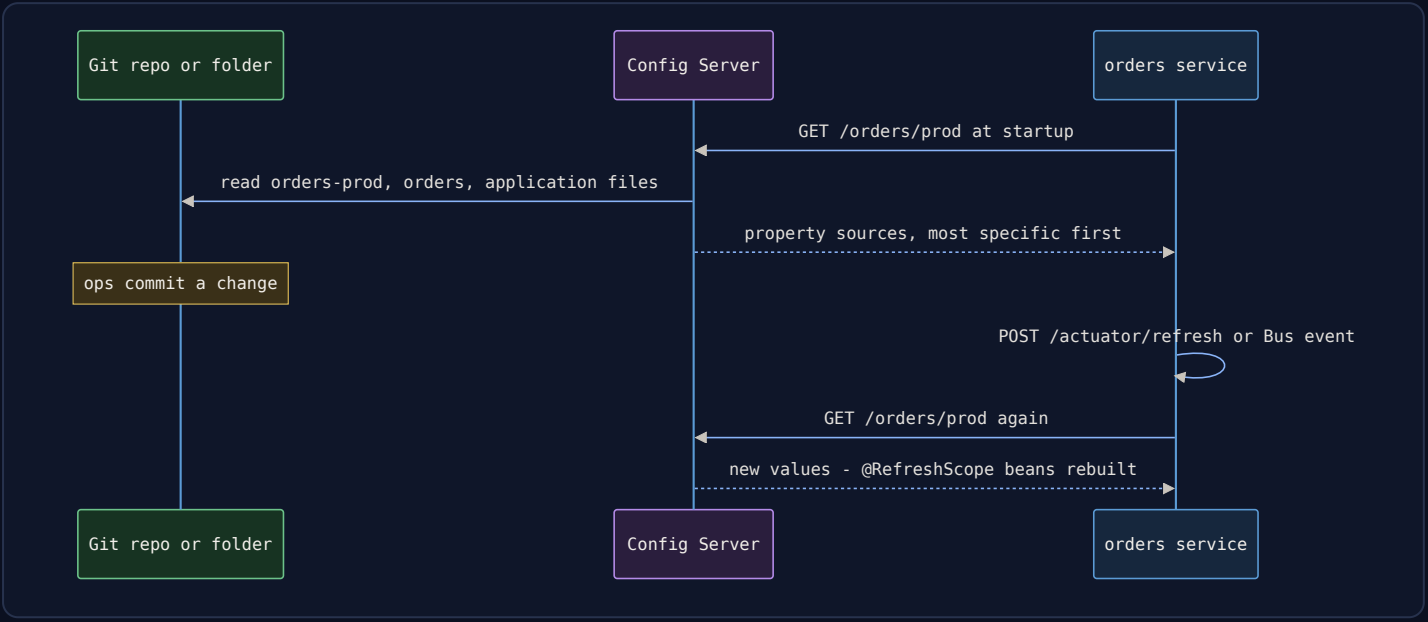
A **Config Server** is a small Boot app serving configuration files over HTTP from a backend — usually a **Git repository**, also a local folder (`native` profile), Vault (HashiCorp's secret store), JDBC, S3 and others. Every service is a **config client** importing its properties at startup, layered by *application name*, *profile* and *label* (Git branch/tag) — like one shared company handbook each department reads (general chapter plus its own) and can re-read when updated.

Why it matters

30 services × 4 environments = 120 configuration sets. A Config Server gives one place to review, version (Git history = audit log) and change them, and to refresh running services without a rebuild. See Config externalization notes.

How it works

1. The server exposes `GET /{application}/{profile}[/{label}]` (plus file-style URLs like `/{application}-{profile}.properties`).
2. For `orders` + profile `prod` it returns an ordered list of property sources — most specific first: `orders-prod.properties` → `orders.properties` → `application-prod.properties` → `application.properties`.
3. The client adds `spring.config.import=configserver:http://config:8888` (Boot's config-data API; `bootstrap.yml` is legacy); `optional:configserver:...` if it may start without it.
4. Remote properties join the `Environment` and **override the service's own** `application.properties`. Whether local system properties or environment variables may override remote values is controlled repository-side (`spring.cloud.config.allow-override`, `override-none`, `override-system-properties`).
5. On refresh the client refetches, rebinds `@ConfigurationProperties` beans and re-creates `@RefreshScope` beans.



The demo

`ConfigServerAndClientTests` writes three files into a temp folder (the "Git repo"), starts a `native` Config Server on a random port, then an `orders` client with profile `prod`:

```
// server
@SpringBootApplication
@EnableConfigServer
public class ConfigServerApplication { ... }
//   --spring.profiles.active=native
//   --spring.cloud.config.server.native.search-locations=file:<repo>/

// client bean
@Component
@RefreshScope   // re-created with fresh values after a refresh
public class OrderSettings {
    public OrderSettings(@Value("${orders.discount-percent}") int discountPercent,
                         @Value("${orders.greeting}") String greeting,
                         @Value("${shared.timeout}") String timeout) { ... }
}
//   --spring.application.name=orders --spring.profiles.active=prod
//   --spring.config.import=configserver:http://localhost:<port>
```

File in the repo	Content
`application.properties` (all services)	`shared.timeout=5s`, `orders.greeting=hello from application.properties`
`orders.properties`	`orders.greeting=hello from orders.properties`, `orders.discount-percent=10`
`orders-prod.properties`	`orders.discount-percent=15`

Then it edits `orders-prod.properties` to 20 and calls `ContextRefresher.refresh()` (what `POST /actuator/refresh` does):

```
GET /orders/prod -> property sources, highest priority first:
   orders-prod.properties
   orders.properties
   application.properties
client sees:    discount=15%, greeting='hello from orders.properties', timeout=5s
refresh changed [orders.discount-percent]
after refresh:  discount=20%, greeting='hello from orders.properties', timeout=5s
```

- `discount=15%` from the profile file, `greeting` from the service file (overriding the shared one), `timeout` from the shared file; `refresh()` returns exactly the changed keys.

Demo-only detail: server and client share one classpath, so the client passes `spring.cloud.config.enabled=true` — the server jar switches the client off by default (`ConfigServerBootstrapApplicationListener`) so a Config Server doesn't fetch from itself. Real deployments are separate apps.

Refreshing many instances, secrets and alternatives

- **Spring Cloud Bus** (`spring-cloud-starter-bus-kafka` / `-amqp`) broadcasts one refresh to all instances (`POST /actuator/busrefresh`); `spring-cloud-config-monitor` can trigger it from a Git webhook (HTTP call from the Git host on push).
- `{cipher}AQB...` values are decrypted by the server (`encrypt.key` or a keystore); better keep secrets in **Vault** (`spring-cloud-starter-vault-config`) or Kubernetes Secrets.
- On Kubernetes, ConfigMaps/Secrets mounted as files and imported with `spring.config.import=configtree:/etc/config/` need no extra server (3.8); Spring Cloud Kubernetes can also read ConfigMaps via the API and reload.

Trade-offs and when not to use it

Pro	Con
One reviewed, versioned source; environment diffs visible	Another critical service: if down, new instances can't start (use `optional:`, `fail-fast` + retry, ≥2 replicas)
Runtime refresh without redeploy	Runtime refresh is also runtime *risk* — a bad value hits all instances at once
Works on any platform	On Kubernetes, ConfigMaps may be enough

Common mistakes

- Expecting every bean to see refreshed values — only `@ConfigurationProperties` (rebound) and `@RefreshScope` (re-created) beans do; a `@Value` field in a normal singleton keeps the old value.
- Plain-text secrets in the Git repo.
- Forgetting `spring.application.name` — the client then asks for `application`, not `orders`.

16.4 Service discovery: Eureka, Consul, Kubernetes

What it is

Instances come and go and their IP addresses change. **Service discovery** keeps a **registry** of live instances per service name; clients ask "where is `inventory`?" instead of hard-coding addresses (Service Discovery notes: client- vs server-side).

Spring Cloud Commons defines `DiscoveryClient` (`getInstances("inventory")` → `ServiceInstance`s with host, port, secure flag, metadata); each backend implements it:

Backend	Starter	How instances register	Health	Notes (not run here)
Netflix Eureka	`spring-cloud-starter-netflix-eureka-server` / `-client`	client registers itself and sends heartbeats (every 30 s by default)	heartbeat lease expiry	AP design (CAP: stays **A**vailable during a network **P**artition rather than **C**onsistent): may serve stale entries (self-preservation mode)
HashiCorp Consul	`spring-cloud-starter-consul-discovery`	client registers via the local Consul agent	Consul calls the app's health URL	also KV store for config (`spring-cloud-starter-consul-config`)
Kubernetes	`spring-cloud-starter-kubernetes-client-all` (or `-fabric8`)	Kubernetes itself (Endpoints of a Service)	readiness probes	needs RBAC (role-based access control) permission to read endpoints
Simple (in Commons)	none	static list in properties	none	tests, local dev, fixed topologies — used in 16.5
Zookeeper	`spring-cloud-starter-zookeeper-discovery`	ephemeral nodes (vanish when the owner disconnects)	session	CP design (consistent; may refuse requests during a partition)

Eureka in a nutshell (not run):

```
@SpringBootApplication
@EnableEurekaServer              // the registry: a dashboard at http://localhost:8761
public class RegistryApplication { ... }
```

```yaml
# each service (client) - with spring-cloud-starter-netflix-eureka-client on the classpath
spring.application.name: inventory                  # the service id
eureka:
  client:
    service-url:
      defaultZone: http://registry:8761/eureka/
  instance:
    prefer-ip-address: true
```

On Kubernetes, do you need it?

Usually **no**: a `Service` named `inventory` gives a stable DNS name (`http://inventory:8080`), kube-proxy spreads connections over ready pods, and readiness probes cover health (server-side discovery and load balancing). Add Spring Cloud Kubernetes only for client-side features: instance metadata, per-request balancing over pods, discovery across namespaces.

16.5 Client-side load balancing with Spring Cloud LoadBalancer

What it is

Client-side load balancing: the *caller* picks an instance from the list discovery provides — no load-balancer box in between (System Design Fundamentals). Spring Cloud LoadBalancer (Netflix Ribbon's replacement) resolves a service id in a URL (`http://inventory/...` or `lb://inventory`) to a concrete instance per request.

How it works

1. A client marked for load balancing (`@LoadBalanced RestClient.Builder` , a Feign client with only a `name` , an HTTP service group without a `base-url` , the gateway's `lb://` URIs) makes a request to `http://inventory/stock/1` .
2. An interceptor asks the `ReactorLoadBalancer` for service `inventory` .
3. A `ServiceInstanceListSupplier` gets instances from the `DiscoveryClient` (cached), optionally filtered (health-check, zone, hints).
4. The algorithm picks one — **round robin** by default (instances in turn: A, B, A, B; randomised start), or `RandomLoadBalancer` , weighted, "same instance preference", subset.
5. The URL is rewritten to `http://host:port/stock/1` and sent.

The demo: static instances, two clients

`service-clients` uses the **simple** discovery client (instances listed in properties). The test starts two tiny HTTP servers, "A" and "B", and points the properties at them:

```properties
spring.application.name=orders
# "Simple" discovery: a fixed list of instances per service id (instead of Eureka/Consul/Kubernetes).
spring.cloud.discovery.client.simple.instances.inventory[0].uri=http://localhost:9001
spring.cloud.discovery.client.simple.instances.inventory[1].uri=http://localhost:9002
# No spring.http.serviceclient.inventory.base-url: Spring Cloud LoadBalancer then uses
# the group name "inventory" as the service id and picks an instance per request.
```

Output (both clients shown in 16.6):

```
HTTP interface 4 calls served by [A, B, A, B]
OpenFeign      4 calls served by [A, B, A, B]  5th call -> Stock[sku=SKU-9, quantity=7, servedBy=A]
```

Round robin alternates; the start is random, so another run may print `[B, A, B, A]` (the test asserts alternation, not order). The startup log warns:

```
WARN ... LoadBalancerCaffeineWarnLogger : Spring Cloud LoadBalancer is currently working with the default cache.
While this cache implementation is useful for development and tests, it's recommended to use Caffeine cache in
pr...
```

In production add `com.github.ben-manes.caffeine:caffeine` (a fast in-memory cache) so the instance-list cache is bounded and refreshed properly (`spring.cloud.loadbalancer.cache.ttl` — time to live, default 35 s).

Useful settings:

```
# instance-list supplier chain: default | health-check | zone-preference | weighted | same-instance-preference | request-based-sticky-session | subset
spring.cloud.loadbalancer.configurations=health-check
# per-service overrides live under spring.cloud.loadbalancer.clients.<serviceId>.*
```

For a plain `RestClient`, mark the builder:

```
@Bean @LoadBalanced
RestClient.Builder loadBalancedBuilder() { return RestClient.builder(); }
// restClientBuilder.baseUrl("http://inventory").build()
```

Trade-offs

Client-side (Spring Cloud LoadBalancer)	Server-side (K8s Service, NGINX, cloud LB, mesh)
per-request choice, custom algorithms, zone/metadata aware	nothing in the app; works for every language
each client needs the library and a discovery source	extra network hop (except kube-proxy/mesh sidecars), fewer app-level decisions

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16.6 Declarative clients: OpenFeign vs HTTP interface clients

What it is

Both turn a Java interface into a generated proxy making the HTTP calls (6.6 covers HTTP interfaces in depth).

```
// Spring Cloud OpenFeign: name = service id -> resolved by the LoadBalancer (no url given)
@FeignClient(name = "inventory")
public interface InventoryFeignClient {
    @GetMapping("/stock/{sku}")                    // Spring MVC annotations
    Stock stock(@PathVariable("sku") String sku);
}

// Spring Framework HTTP service interface (no extra library)
public interface InventoryHttpClient {
    @GetExchange("/stock/{sku}")
    Stock stock(@PathVariable String sku);
}

@SpringBootApplication
@EnableFeignClients                                                    // scan for @FeignClient
@ImportHttpServices(group = "inventory", types = InventoryHttpClient.class) // group name = service id
public class ClientsApplication { ... }
```

Since Spring Cloud 2025.1 (Commons 5.0), LoadBalancer plugs into Boot 4's **HTTP service client groups**: a group without `spring.http.serviceclient.<group>.base-url` gets a load-balanced `http://<groupName>`, and an `lb://` base URL also works — so `inventory` needed no URL. `LoadBalancedClientsTests` calls each client four times (output in 16.5).

Comparison

	OpenFeign	HTTP interface client
Where it lives	Spring Cloud OpenFeign (separate project)	Spring Framework 6+ core; Boot 4 groups via `@ImportHttpServices`
Status	**feature-complete / maintenance** — only fixes	actively developed; the Spring team's recommendation
Annotations	Spring MVC's `@GetMapping`, ... on the interface	`@GetExchange`, `@PostExchange`, ...
Underlying client	Feign's own client (or Apache HC5 / OkHttp)	`RestClient` or `WebClient` (reactive return types possible)
Config	`spring.cloud.openfeign.client.config.<name>.*` (timeouts, loggers, interceptors)	`spring.http.serviceclient.<group>.*`, `RestClientHttpServiceGroupConfigurer`
Load balancing	automatic when only `name` is set	automatic for groups without base URL (Spring Cloud 2025.1+)
Circuit breaker	`spring.cloud.openfeign.circuitbreaker.enabled=true` + fallbacks	Spring Cloud 2025.1+: per-group breaker with an `@HttpServiceFallback` class (switch `spring.cloud.circuitbreaker.http-services.enabled`, on by default; not run here), or wrap the call in a service method with Resilience4j
Tracing	instrumented	instrumented (via the observed `RestClient`)

Recommendation: new code → HTTP interface clients; existing Feign code keeps working — migrate when convenient (annotations map almost one-to-one).

16.7 API gateway: Spring Cloud Gateway

What it is

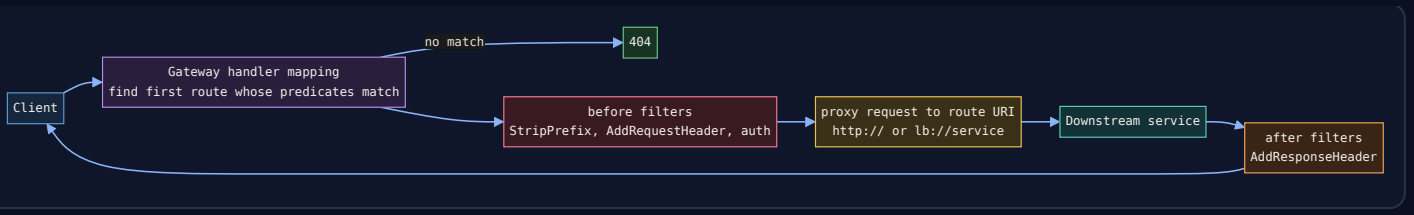
An **API gateway** is the single front door for external clients: it **routes** each request to the right service and applies **edge concerns** once — authentication/token relay, rate limiting, CORS, header rewriting, retries/circuit breaking, request size limits. Patterns: API Gateway notes, BFF, API & Edge module.

Two flavours, same concepts:

	Server Web MVC	Server WebFlux
Starter	`spring-cloud-starter-gateway-server-webmvc`	`spring-cloud-starter-gateway-server-webflux`
Runtime	Servlet (Tomcat/Jetty), blocking; pairs well with virtual threads	Netty, reactive
Routes in code	`RouterFunction` DSL (`GatewayRouterFunctions.route`)	`RouteLocatorBuilder` DSL
Properties	`spring.cloud.gateway.server.webmvc.routes`	`spring.cloud.gateway.server.webflux.routes`
Choose when	team knows MVC, uses blocking libraries	very high connection counts, streaming, already reactive

How it works

A **route** = id + destination URI + **predicates** (when does it match? `Path`, `Method`, `Header`, `Host`, `Query`, `Cookie`, `After`, `Weight`, `Version` ...) + **filters** (what to change? `StripPrefix`, `AddRequestHeader`, `RewritePath`, `CircuitBreaker`, `Retry`, `RequestRateLimiter`, `TokenRelay`, ...).



The demo

`gateway-demo` defines one route in YAML and one in Java.

```yaml
spring:
  cloud:
    gateway:
      server:
        webmvc:
          routes:
            # Route 1, in YAML: /api/orders/** -> orders service
            - id: orders
              uri: ${services.orders.url}              # with discovery: lb://orders
              predicates:
                - Path=/api/orders/**
                - Method=GET,POST
              filters:
                - StripPrefix=1                          # /api/orders/7 -> /orders/7
                - AddRequestHeader=X-Gateway, ch16-gateway
                - AddResponseHeader=X-Served-Via, gateway
```

```java
/** Route 2, in the Java DSL: /api/inventory/** -> inventory service, protected by a circuit breaker. */
@Bean
RouterFunction<ServerResponse> inventoryRoute(@Value("${services.inventory.url}") String inventoryUrl) {
    return route("inventory")
            .GET("/api/inventory/**", http())
            .before(uri(inventoryUrl))
            .filter(circuitBreaker("inventoryCB", URI.create("forward:/fallback/inventory")))
            .build();
}

@GetMapping("/fallback/inventory")
String inventoryFallback() {
    return "{\"items\":[],\"note\":\"inventory temporarily unavailable\"}";
}
```

(Static imports: `GatewayRouterFunctions.route`, `HandlerFunctions.http`, `BeforeFilterFunctions.uri`, `CircuitBreakerFilterFunctions.circuitBreaker`. Pass the target via `before(uri(...))`: the `http(String)` / `http(URI)` overloads, deprecated in Gateway 4.x, are gone in Gateway 5.0 — only `http()` remains.) The `CircuitBreaker` filter uses Spring Cloud CircuitBreaker, hence `spring-cloud-starter-circuitbreaker-resilience4j`.

`GatewayRoutingTests` runs an echoing "orders service" and points the inventory route at a port where nothing listens:

```
GET /api/orders/7 -> orders-service saw GET /orders/7?expand=items X-Gateway=ch16-gateway | response X-Served-Via=gateway
DELETE /api/orders/7 and GET /api/unknown -> 404 (no route matches)
GET /api/inventory/items (service down) -> {"items":[],"note":"inventory temporarily unavailable"}
```

- `StripPrefix=1` removed `/api`; the query string was kept; request and response headers were added.
- `DELETE` fails the `Method` predicate and nothing matches `/api/unknown` → 404.
- The inventory connection failure went through the circuit breaker to the fallback — a degraded but valid answer.

Other common gateway jobs (not run)

- **Token relay** (`TokenRelay` filter + `spring-boot-starter-oauth2-client`): the gateway logs the user in and forwards the access token — 11.10, BFF in 11.12.
- **Rate limiting**: the WebFlux flavour's `RequestRateLimiter` (Redis); the MVC flavour's Bucket4j-based filter. See Rate Limiting.
- **Weighted routes** (`Weight=group, 90`) for canary releases (a small traffic share to a new version first).

Common mistakes

- Business logic or many-service aggregation in the gateway → a "distributed monolith" bottleneck. Keep it thin; use a BFF for client-specific aggregation.
- Spring Cloud Gateway *and* an Ingress/cloud gateway doing the same things — one layer per concern.
- Retrying non-idempotent `POST`s at the gateway (an order-creating `POST` can't safely run twice).

16.8 Resilience: circuit breaker, retry, rate limiter

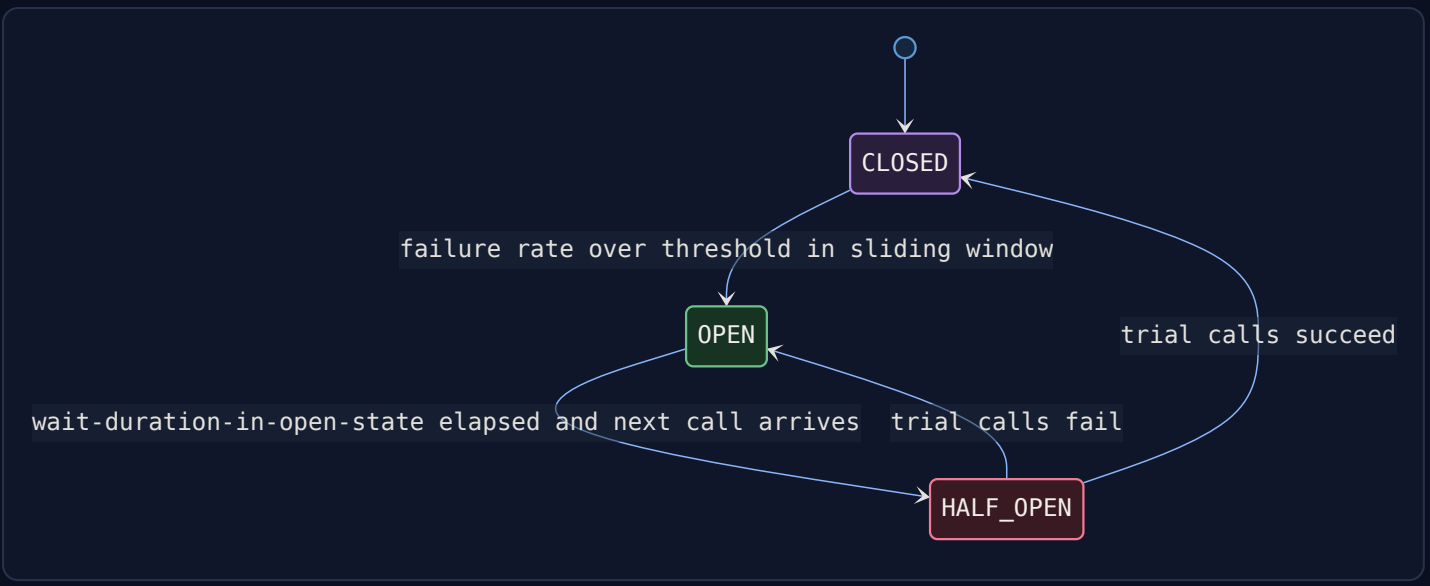
What it is

Remote calls fail and, worse, become **slow**; unprotected callers pile up waiting threads until they fail too — a **cascading failure**. Tools (Reliability module):

Tool	Idea	Notes
Timeout	never wait forever	the first line of defence; set on every client (Chapter 6)
Retry	try again after a transient failure, with back-off (growing waits) + jitter (random extra wait so clients don't retry in lockstep)	only idempotent operations — Retry notes
Circuit breaker	after many failures, stop calling for a while and fail fast / fall back	Circuit Breaker notes
Bulkhead	cap concurrent calls to one dependency	Bulkhead notes
Rate limiter	cap calls per time period	protects a dependency or enforces a quota
Fallback	a degraded answer (cache, default, empty list)	Graceful degradation

Resilience4j is the standard Java library for all of these (Hystrix's successor): directly (`resilience4j-spring-boot4` : annotations + properties + Micrometer metrics + health) or via **Spring Cloud CircuitBreaker**, an abstraction (`CircuitBreakerFactory.create("id").run(supplier, fallback)`) with a Resilience4j implementation, used e.g. by the gateway filter above.

Circuit breaker states



- **CLOSED** — calls pass; outcomes are recorded in a **sliding window** (last N calls, or last N seconds).
- **OPEN** — calls are rejected immediately with `CallNotPermittedException` (→ fallback) without touching the dependency, giving it time to recover.
- **HALF_OPEN** — a limited number of trial calls decide whether to close again.

The demo

`resilience-demo` wraps a fake remote `RemoteInventory` (the tests switch it up or down):

```
@CircuitBreaker(name = "inventory", fallbackMethod = "cachedStock")
public String stockWithBreaker(String sku) {
    return "live stock " + remote.stock(sku);
}

// fallback: same parameters + the exception as the last parameter
private String cachedStock(String sku, Throwable t) {
    return "cached stock 5 (" + t.getClass().getSimpleName() + ")";
}

@Retry(name = "inventory-retry")
public String stockWithRetry(String sku) { return "live stock " + remote.stock(sku); }

@RateLimiter(name = "search")
public String search(String text) { return "results for " + text; }
```

```yaml
resilience4j:
  circuitbreaker:
    instances:
      inventory:
        sliding-window-type: COUNT_BASED
        sliding-window-size: 4                # judge the last 4 calls
        minimum-number-of-calls: 4            # ... but only once 4 calls were recorded
        failure-rate-threshold: 50            # open when >= 50 % of them failed
        wait-duration-in-open-state: 300ms    # stay OPEN this long, then allow trial calls
        permitted-number-of-calls-in-half-open-state: 2
        record-exceptions: ch16.resilience.RemoteCallException
  retry:
    instances:
      inventory-retry:
        max-attempts: 3                       # total attempts, including the first call
        wait-duration: 20ms
        enable-exponential-backoff: true
        exponential-backoff-multiplier: 2     # 20 ms, then 40 ms
        retry-exceptions: ch16.resilience.RemoteCallException
  ratelimiter:
    instances:
      search:
        limit-for-period: 3                   # 3 permits ...
        limit-refresh-period: 10s             # ... per 10-second cycle
        timeout-duration: 0                   # don't wait for a permit: reject immediately
```

`CircuitBreakerTests` subscribes to the breaker's state-transition events and walks it through all states:

```
1) dependency goes down, 4 calls:
   call 1: cached stock 5 (RemoteCallException)  state=CLOSED
   call 2: cached stock 5 (RemoteCallException)  state=CLOSED
   call 3: cached stock 5 (RemoteCallException)  state=CLOSED
   call 4: cached stock 5 (RemoteCallException)  state=OPEN
2) while OPEN the remote is not even called:
   call 5: cached stock 5 (CallNotPermittedException)  remote calls +0
3) wait 300 ms, dependency is back; 2 trial calls in HALF_OPEN:
   call 6: live stock 7  state=HALF_OPEN
   call 7: live stock 7  state=CLOSED
transitions: [CLOSED -> OPEN, OPEN -> HALF_OPEN, HALF_OPEN -> CLOSED]
```

- Calls 1–3 fail but the breaker stays CLOSED: the window needs `minimum-number-of-calls` (4) first.
- After call 4 the failure rate is 100% ≥ 50% → OPEN.
- Call 5 is rejected without calling the remote (`remote calls +0`) — what stops the cascade.
- After 300 ms the next call moves it to HALF_OPEN — by default (`automatic-transition-from-open-to-half-open-enabled: false`) no timer does this; the first call after the wait does. Two successful trials close it.
- Resilience4j defaults otherwise: `COUNT_BASED` window of 100 calls, `minimum-number-of-calls` 100, `failure-rate-threshold` 50, `wait-duration-in-open-state` 60 s, 10 permitted HALF_OPEN calls — sensible in production, too slow for a demo. Retry defaults to 3 attempts 500 ms apart.

`RetryAndRateLimiterTests` :

```
Framework 7 @Retryable, remote fails twice -> live stock 7 after 3 attempts
RateLimiter (3 per 10 s): call1=OK call2=OK call3=OK call4=REJECTED call5=REJECTED
Resilience4j @Retry, remote fails twice:
   retry #1 after inventory unavailable, waiting 20 ms
   retry #2 after inventory unavailable, waiting 40 ms
   -> live stock 7 after 3 attempts
   retry #1 after inventory unavailable, waiting 20 ms
   retry #2 after inventory unavailable, waiting 40 ms
   remote down: gave up after 3 attempts, caller gets RemoteCallException
```

- Resilience4j's `max-attempts: 3` counts the **first call**; Framework 7's `maxRetries = 2` counts only **retries** — both mean 3 attempts (`maxRetries = 3` would be 4).
- Exponential back-off doubled the wait (20 → 40 ms); add jitter in production (`randomized-wait-factor` / `enable-randomized-wait`).

- The rate limiter rejects with `RequestNotPermitted` once the period's 3 permits are used.

The Framework 7 version (Chapter 4.7 covers it fully):

```
@Retryable(includes = RemoteCallException.class, maxRetries = 2, delay = 20, multiplier = 2)
@ConcurrencyLimit(limit = 10, policy = ConcurrencyLimit.ThrottlePolicy.REJECT)   // a simple bulkhead
public String stock(String sku) { return "live stock " + remote.stock(sku); }
// needs @EnableResilientMethods on a configuration class
```

Resilience4j vs Framework 7 core

Need	Framework 7 core	Resilience4j
Retry with back-off, jitter	`@Retryable`, `RetryTemplate`	`@Retry`, `Retry`
Bulkhead (concurrency cap)	`@ConcurrencyLimit` (BLOCK or REJECT)	`@Bulkhead` (semaphore or thread pool)
Circuit breaker	—	`@CircuitBreaker`
Rate limiter	—	`@RateLimiter`
Time limiter	— (use client timeouts)	`@TimeLimiter` (async/reactive)
Fallback methods	— (catch at the caller)	`fallbackMethod`
Config in properties, per instance	attributes (placeholders allowed)	yes, rich
Metrics, health, events	`MethodRetryEvent`	Micrometer meters, health indicators, event publishers
Extra dependency	none	yes

Rule of thumb: plain retries and concurrency caps → Framework 7; anything involving circuit breaking, rate limiting or fallbacks → Resilience4j.

Order when stacking annotations (Resilience4j defaults): `Retry( CircuitBreaker( RateLimiter( TimeLimiter( Bulkhead( call ) ) ) ) )` — retry outermost, so each attempt is counted by the breaker, and an OPEN breaker's rejection can be retried unless you exclude `CallNotPermittedException`. Reorder with `resilience4j.*.<type>-aspect-order`.

Common mistakes

- Retrying everything, including 4xx and non-idempotent calls.
- Retries at gateway **and** client **and** service → retry storms (3 × 3 × 3 = 27 calls).
- A breaker without timeouts: slow calls never "fail", so it never opens (or configure `slow-call-duration-threshold` / `slow-call-rate-threshold`).
- One breaker for all endpoints of a service — one bad endpoint opens the circuit for healthy ones.
- Self-invocation — all these annotations are proxy-based (Chapter 4).
- Fallbacks hiding outages forever — alert on breaker state (`resilience4j_circuitbreaker_state` metric).

TL;DR

Timeouts first; retry only transient, idempotent failures with back-off + jitter; a circuit breaker to fail fast and give the dependency room; bulkhead/rate limiter to cap load; fallbacks for degraded answers. Framework 7 covers retry + concurrency limits; Resilience4j covers everything.

16.9 Distributed tracing across services

Little beyond Chapter 15.7: every service adds `spring-boot-starter-opentelemetry` and exports to the same collector; the gateway, clients from Boot's `RestClient.Builder` (including HTTP interface groups) and OpenFeign propagate `traceparent`. For messaging enable observation explicitly

(`spring.kafka.template.observation-enabled=true`, `spring.kafka.listener.observation-enabled=true`, likewise RabbitMQ) so context travels in message headers. Microservice-specific points:

- Use the **same propagation format** everywhere (W3C by default; set `management.tracing.propagation.type=b3` only for legacy Zipkin/B3 services, or produce both).
- Set `spring.application.name` in every service — it becomes the service name in traces and logs.
- Let the **edge decide sampling** (the gateway) — downstream services follow the sampled flag.
- Tag spans with business ids as **high-cardinality** values (order id), never as metric tags.

16.10 Contract testing

What it is

Consumer A and provider B are tested separately — do they still agree on the API? End-to-end tests of all services are slow and flaky. **Contract testing** checks each side against a shared **contract** (request → expected response):

1. Consumer-driven: the consumer writes what it needs.
2. The provider's build **verifies** it responds that way.
3. The consumer's tests run against a **stub** (fake server with canned answers) generated from the contract.

Tool	How	Status
Pact (pact-jvm)	consumer tests produce a pact file; a Pact Broker (stores contracts and results) shares it; provider verifies; `can-i-deploy` gate	active, polyglot
Spring Cloud Contract	contracts in YAML/Groovy/Java in the provider repo; the plugin generates provider tests and a stubs jar; consumers use `@AutoConfigureStubRunner`	JVM/Spring-centric; not in the 2025.1.3 BOM, repository archived in `spring-attic` (see 16.2)
OpenAPI-based checks	validate both sides against the OpenAPI document	good for provider-driven public APIs

Spring Cloud Contract, for recognition (not run):

```
# provider: src/test/resources/contracts/stock/shouldReturnStock.yml
request:
  method: GET
  url: /stock/SKU-1
response:
  status: 200
  headers: { Content-Type: application/json }
  body: { sku: "SKU-1", quantity: 7 }
```

```
// consumer test: runs against a WireMock stub built from the provider's contracts
@SpringBootTest
@AutoConfigureStubRunner(ids = "com.acme:inventory:+:stubs:8090", stubsMode =
StubRunnerProperties.StubsMode.LOCAL)
class InventoryClientContractTest { /* call http://localhost:8090/stock/SKU-1 */ }
```

When not to use it: a single team owning both sides that deploys them together, or public APIs where you can't know consumers (use versioning + OpenAPI instead).

16.11 Messaging between services

Synchronous HTTP chains couple availability: any service down fails the request. Events decouple them in time (Event-Driven notes). Spring options:

- **Spring for Apache Kafka / Spring AMQP** — templates and `@KafkaListener` / `@RabbitListener` (Chapter 14, incl. the transactional outbox).
- **Spring Cloud Stream** — a binder abstraction: `java.util.function` beans (`Supplier` , `Function` , `Consumer`) bound to Kafka/RabbitMQ destinations by configuration (`spring.cloud.function.definition=enrichOrder` , `spring.cloud.stream.bindings.enrichOrder-in-0.destination=orders`). Good for many similar pipelines; hides broker-specific features you may need.
- Cross-service consistency without distributed transactions: Saga (choreography — services react to each other's events — or orchestration — a coordinator directs each step), idempotent consumers, and CQRS read models.

16.12 Common mistakes (whole chapter)

Symptom	Cause	Fix
App fails at startup with a compatibility error	Spring Cloud train doesn't match Boot	2025.1.x for Boot 4.0/4.1
`No spring.config.import property has been defined`	config client on classpath without import	add `spring.config.import=configserver:...` (or `optional:`)
Config change not visible after refresh	plain `@Value` in a singleton	`@RefreshScope` or `@ConfigurationProperties`
`UnknownHostException: inventory`	client not load-balanced	`@LoadBalanced` builder, Feign `name` , or HTTP group without base URL
LoadBalancer "default cache" warning	Caffeine missing	add Caffeine
Gateway returns 404	predicates don't match (method, path, prefix)	check the route definition; DEBUG logging for `org.springframework.cloud.gateway`
Breaker never opens	no timeouts, slow calls not recorded, window too large	timeouts + slow-call thresholds + sensible window
Retry storm during an outage	retries at several layers	retry in one layer, with jitter; breaker around retries
Everything down when one service is down	synchronous chains, no fallbacks	timeouts, breakers, fallbacks, async events

---

16.13 In the interview

Q: Which Spring Cloud version do you use with Spring Boot 4.1? The 2025.1.x "Oakwood" train (Boot 4.0 and 4.1, the latter from 2025.1.2), imported as the `spring-cloud-dependencies` BOM; its components are 5.0.x.

Q: How do microservices find each other in Spring? Through the `DiscoveryClient` abstraction backed by Eureka, Consul, Zookeeper or Kubernetes; Spring Cloud LoadBalancer resolves service ids (`http://inventory` , `lb://inventory`) to instances per request, round robin by default. On Kubernetes plain Services + DNS usually suffice.

- *Client-side vs server-side load balancing?* → The client picks the instance (smarter, per-request, needs a library) vs a proxy picks it (language-agnostic, extra hop).

Q: Explain the circuit breaker states and how you configure them in Resilience4j. CLOSED records outcomes in a sliding window; when the failure (or slow-call) rate exceeds the threshold after `minimum-number-of-calls`, it goes OPEN and rejects calls fast; after `wait-duration-in-open-state` it goes HALF_OPEN and lets `permitted-number-of-calls-in-half-open-state` trial calls decide between CLOSED and OPEN. Configure per instance under `resilience4j.circuitbreaker.instances.<name>`; use `@CircuitBreaker(name, fallbackMethod)`.

- *Different from retry?* → Retry repeats one call hoping for success; the breaker stops calling an obviously unhealthy dependency. Combine them, the breaker protecting against retry storms.
- *Does Framework 7 have a circuit breaker?* → No — only `@Retryable` and `@ConcurrencyLimit`.

Q: What does an API gateway do, and which Spring Cloud Gateway flavour would you pick? Single entry point: routing by predicates, filters for auth/token relay, rate limiting, header rewriting, resilience. Web MVC flavour for servlet teams and virtual threads; WebFlux flavour for reactive stacks and very high concurrency. Keep business logic out.

Q: How do you manage configuration across 30 services? Externalised config: Spring Cloud Config (Git-backed, layered by app/profile/label, refresh via actuator or Bus) or platform-native ConfigMaps/Secrets with `configtree:`; secrets in Vault; config changes reviewed like code.

Q: OpenFeign or HTTP interfaces? → HTTP interfaces for new code (core Spring, actively developed, Boot 4 groups, load-balanced by Spring Cloud); OpenFeign is in maintenance mode.

Q: How do you test that services stay compatible? → Consumer-driven contract tests (Pact, or Spring Cloud Contract in existing projects) plus a small number of end-to-end smoke tests.

Chapter TL;DR

- Spring Cloud = ready-made answers to distributed-system problems; check what your platform (Kubernetes) already gives you before adding them. Patterns: microservices-patterns notes.
- **Boot 4.1 → Spring Cloud 2025.1.x (Oakwood)**, BOM `spring-cloud-dependencies` (2025.1.3 used here); components 5.0.x; OpenFeign in maintenance; Spring Cloud Contract no longer in the 2025.1.3 BOM.
- **Config Server**: `@EnableConfigServer`, Git or `native` backend; clients use `spring.config.import=configserver:`; precedence `app-profile` > `app` > `application`; refresh re-creates `@RefreshScope` beans (proved: 15% → 20%).
- **Discovery**: `DiscoveryClient` over Eureka/Consul/Kubernetes/simple; **LoadBalancer** resolves service ids per request, round robin by default (proved: B, A, B, A); add Caffeine.
- **Declarative clients**: prefer HTTP interfaces (`@ImportHttpServices`, load-balanced when the group has no base URL); OpenFeign still works.
- **Gateway**: routes = predicates + filters; server-webmvc or server-webflux; `StripPrefix`, headers, `CircuitBreaker` with fallback (proved).
- **Resilience4j**: breaker CLOSED → OPEN → HALF_OPEN → CLOSED (proved), retry with back-off, rate limiter (proved); Framework 7 `@Retryable` / `@ConcurrencyLimit` for simple cases.
- Tracing across services = Chapter 15 setup in every service; contracts with Pact/Spring Cloud Contract; events with Kafka/Stream.
- Next: Chapter 17 — Spring AI and what's new.